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The influence of residual stresses on laser beam welding processes of aluminium sheets

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Abstract

Forming processes may lead to residual stresses remaining in the parts. Laser beam welding leads to a local release of the present stresses, which finally results into strain rates and causes the formation of hot cracks. The relation between the acting stresses and the emerging strain rates was investigated experimentally by means of high-speed digital image correlation during welding of aluminium sheets. In the paper an analytical model is presented which describes the influence of the acting stresses on the emerging strain rates and enables the optimization of the process parameters with regard to a prevention of hot crack formation.

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1. Introduction

Hot cracks are a fatal defect in laser beam welding of aluminium alloys. They form during solidification by the rupture of the liquid melt between the solidifying grains. Fig. 1 shows the formation of a centreline crack during laser beam welding, which grows along a continuous grain boundary along the centreline of the weld [1]. The ongoing propagation process of the crack is located in the solidification zone of the weld behind the progressing melt pool.

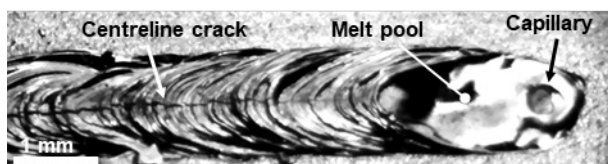


Fig. 1. Frame of a high speed video of the formation of a centreline crack during welding of AlMgSi-alloy of the 6000 series.

The liquid melt at the grain boundaries tears if the strain rate that affects the solidification zone of the weld exceeds a critical

value [2]. This critical value is mainly influenced by the chemical composition of the alloy and by the process parameters [3]. Aluminium alloys of the 6000 series require sufficient compressive strain rates to avoid the formation of hot cracks [3,4]. For the formation of the centreline hot cracks shown in Fig. 1, the strain rates acting transversely to the feed direction are particularly critical.

Deep drawn sheet metal components often contain residual stresses [5]. It was shown in former work [6], that these residual stresses do influence the hot crack formation during laser beam welding of such components. It can be assumed that the release of the residual stresses increases the strain rates, which affect the solidification zone, and thus favours the formation of hot cracks. The present work suggests an analytical model for the calculation of the emerging strain rates by a description of the release of the present stresses under the consideration of the influence of the weld pool geometry and the welding parameters.

2. Analytical Description of the Release of the Stresses

Fig. 2 depicts a sketch of the laser welding process. The melt pool is represented by the red area and the solidification zone is located between the liquidus isotherm (dashed black line) and the solidus isotherm (blue line). The purple arrows represent acting uniaxial stresses σ and the yellow arrows represent the strain rate $\dot{\varepsilon}$ which affects the solidification zone. Only stresses and strain rates which act transverse to the feed direction are considered in the following. The feed direction of the laser beam relative to the sample corresponds to the positive x direction. The coordinates have their origin in the capillary. In case of full penetration laser beam welding the temperature field can be described by two dimensional heat conduction of a moving line source, which is located at $x = 0$ and $y = 0$.

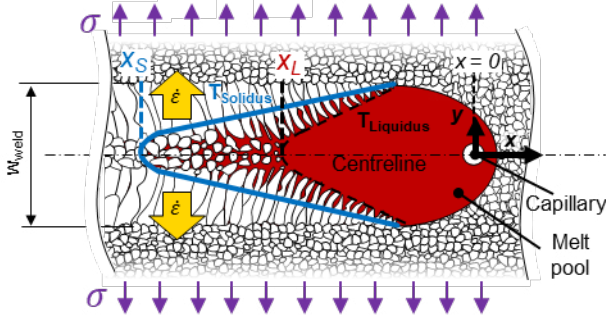


Fig. 2. Sketch of the melt pool and the solidification zone at the rear side of the melt pool during laser beam welding.

As derived from the energy balance of the weld [3,7] the width of the weld

$$w_{weld}(E_{Depth}) = \frac{E_{Depth} \cdot \eta_{abs} \cdot \eta_T}{\rho \cdot (c_p(T_{Liq} - T_{amb}) + h_s)} \quad (1)$$

can be written as a function of the depth specific line energy $E_{Depth} = \frac{P}{v \cdot s}$, which describes the laser power P per welding depth s per welding velocity v , for given liquidus temperature T_{Liq} , ambient temperature T_{amb} , and the material characteristics density ρ , heat capacity c_p , and the latent heat of fusion h_s . The efficiency of the process is described by the product of the heat efficiency η_T and the absorptance of the laser beam in the sample η_{abs} . In case of full penetration welding, the welding depth s equals the initial sheet thickness s_0 .

For the description of centreline hot cracking, it is sufficient to describe the processes at the centreline at $y = 0$ between $x = x_L$ and $x = x_S$ where the solidification takes place, as shown in [3,7]. According to [7], the cooling rate

$$\dot{T}(E_{Depth}) = \frac{2 \cdot (T_{Liq} - T_{amb})^3 \cdot \lambda_{th} \cdot \rho \cdot c_p \cdot \pi}{\eta_{abs}^2 \cdot E_{Depth}^2} \quad (2)$$

at the centreline ($y = 0$) at the liquidus isotherm T_{Liq} ($x = x_L$) is also a function of the absorbed depth specific line energy $\eta_{abs} \cdot E_{Depth}$, where λ_{th} is the heat conductivity of the welded material. The multiplication of the cooling rate $\dot{T}(E_{Depth})$ with the thermal expansion coefficient α yields the strain rate

$$\dot{\varepsilon}_{th}(E_{Depth}) = -\dot{T}(E_{Depth}) \cdot \alpha, \quad (3)$$

which affects the solidification zone at $x = x_L$.

Under the assumption of uniaxial elastic deformation, the simple relation $\varepsilon = \frac{\sigma}{E}$ describes the transverse strain ε in case of a present (residual) transverse stress σ in a material with the

Young's modulus E . In case of welding a sheet with a present residual stress σ , the melt pool interrupts the solid material continuity over the length of its width w_{weld} and the present stress releases as strain. This release of stresses takes place at the velocity v of the welding process, that results in a strain rate which affects the solidification zone in addition to the strain rate $\dot{\varepsilon}_{th}$ originating from equation (3). Therefore, the total strain rate

$$\dot{\varepsilon}(\sigma, v, E_{Depth}) = \frac{\sigma \cdot v}{E \cdot w_{weld}(E_{Depth})} + \dot{\varepsilon}_{th}(E_{Depth}), \quad (4)$$

which affects the rear end of the weld pool, is a function of the depth specific line energy E_{Depth} and proportional to the welding velocity v as well as to the present stresses σ .

3. Experimental Simulation of Residual Stresses

Fig. 3 depicts the experimental setup, which is based on the experiments presented in [8]. The diagram depicts the position of the focusing optics and the high-speed camera as well as the clamping of the samples.

In order to obtain a generally valid investigation, the shape constraint of formed parts was simulated experimentally by a uniaxial static elongation of the sample (brown), which caused a uniaxial tensile stress before the welding process started. For this purpose, the clamping device used in the experiments includes a force sensor (yellow) and provides the possibility to stretch the sample to a static elongation, as indicated by the blue arrows.

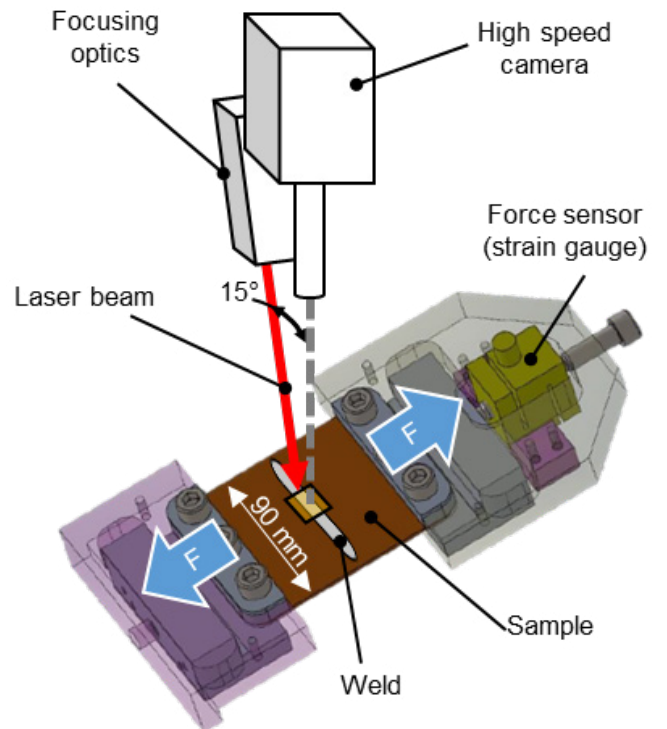


Fig. 3. Experimental setup.

The samples (brown) investigated were sheets of the AA6014 aluminium alloy Ac170 from Novelis having a thickness of $s_0 = 1$ mm and a width of $b_0 = 90$ mm.

Before the process started, the sample (brown) was loaded by an initial external force F_0 (blue arrows) in order to achieve a static elongation during the process. The initial cross section $A_0 = 90 \text{ mm}^2$ which is exposed to the initial force F_0 results

from the multiplication $b_0 \cdot s_0$ and the initial stress taken from the ratio $\sigma_0 = \frac{F_0}{A_0}$.

The release of the stresses during welding was measured by the decrease of the force $F(t)$ during the process by the strain gauge (yellow) at a frequency of 1 kHz. In case of welding without hot crack formation it was assumed, that the cross section $A(t)$ equals approximately to the constant value of the initial cross section $A_0 = 90 \text{ mm}^2$ during the process. The ratio between the measured force $F(t)$ and the constant area A_0 yields the transient determination of the present stress

$$\sigma(t) = \frac{F(t)}{A_0}. \quad (5)$$

As shown in Fig. 1, the growth of centreline hot cracks follows the progressing melt pool with the same velocity and after the first initiation of a crack the stressed sectional area

$$A(t) = s_0 \cdot (b_0 - l_c(t)) \quad (6)$$

decreases with increasing crack growth, i.e. increasing crack length $l_c(t)$. In case of a present centreline crack the acting stress

$$\sigma(t) = \frac{F(t)}{A(t)} \quad (7)$$

results from the ratio between the actual measured force $F(t)$ and the remaining stressed area $A(t)$.

The emerging strain rates $\dot{\epsilon}(t)$, which affect the end of the melt pool at $x = x_L$ were determined during the process by means of digital image correlation of the high speed videos of the welding process, as described in [8].

All experiments were performed with a TruDisk 8001 laser from TRUMPF. The beam was guided by a fibre with a core diameter of 100 μm and a numerical aperture of 0.1 rad yielded a beam quality of $M^2 \approx 15$. The magnification of the processing optics was 2.8, resulting in a focal diameter of $d_f = 280 \mu\text{m}$ and a Rayleigh length of $z_R = 3.92 \text{ mm}$. The moving beam was inclined opposite to feed direction by 15° with respect to the normal of the sample surface and focused onto the surface of the samples.

In order to evaluate the influence of the velocity v and the depth specific Line energy E_{Depth} , as stated by equation (1)-(4), these quantities were varied as listed in Table 1 for the welding parameters applied in the experiments.

Table 1. Welding parameters.

Parameter	Setting 1	Setting 2
Power P	1200 W	1800 W
Velocity v	3 m/min	6 m/min
Depth specific Line energy E_{Depth}	24 J/mm ²	18 J/mm ²
Spot diameter d_f	280 μm	280 μm

4. Results

The equations (1)-(4) were calculated based on the material properties of an AA6014 aluminium alloy as listed in Table 2 for the two welding parameter settings listed in Table 1. The absorptance $\eta_{\text{abs}} \approx \text{const.} = 0.53$ of the laser beam was calculated according to [9] for a conically shaped vapour capillary with an aspect ratio of $\frac{s}{d_f} \approx 3.6$ in liquid aluminium with an absorptivity of 11% at a wavelength of 1.03 μm [10]. The heat

efficiency $\eta_T = 0.25$ was chosen according to Rykalin's calculations for a typical Peclet number of $\text{Pe} = \frac{w_{\text{weld}} \cdot v}{4 \cdot \kappa} \approx 0.2$ [11].

Table 2. Material properties of AA6014 [12] and process efficiencies.

Material parameter	Abbr.	Value	Unit
Thermal conductivity	λ_{th}	211	W/mK
Thermal diffusivity	κ	87.2	$10^{-6} \text{ m}^2/\text{s}$
Heat capacity	c_p	896	J/kgK
Density	ρ	2700	kg/m ³
Latent heat of fusion	h_s	390	kJ/kg
Young's modulus	E	70	GPa
Absorptance	η_{abs}	0.53	1
Heat efficiency	η_T	0.25	1

Fig. 4 shows the calculated strain rates as a function of the acting uniaxial tensile stress. The red dashed line represents the calculated strain rates for the parameter setting 1 and the blue dashed line represents the calculated strain rates for the parameter setting 2, as listed in Table 1.

In order to validate the model, the results of the calculations are compared with the experimental results. For this purpose, the measured strain rate values at $x = x_L$ were correlated with the temporally corresponding results of the stress determination according to equation (5) or (7). The data points in Fig. 4 represent the average strain rate and stress values of minimum five measurements in five different welds. The red data points represent the experimental results in case of welding with parameter setting 1 whereas the blue data points represent the experimental results in case of welding with parameter setting 2. The lengths of the error bars indicate the range between the minimum and maximum of the measured values.

The data points of the experiments in which a hot crack was formed are marked by a black cross. The data points without a black cross represent the experiments that did not show the formation of hot cracks during welding. The calculated results of the model agree well with the experimental results, if hot cracks are absent. Both the experimental and the calculated results show a linear increase of the emerging strain rates with increasing acting stresses.

If hot crack formation occurs, the experimental results differ strongly from the calculated results. As shown in [8], the separation of the material between the two sides resulting from a present hot crack strongly changes the mechanical and thermal boundary conditions. The presented model cannot describe such changes, as it is based on the two-dimensional heat conduction equation of [13], which is limited to an ideal indefinite sheet. Centreline hot cracks appeared for stresses beyond 100 MPa or in case of emerging strain rates that are higher than about -7 %/s.

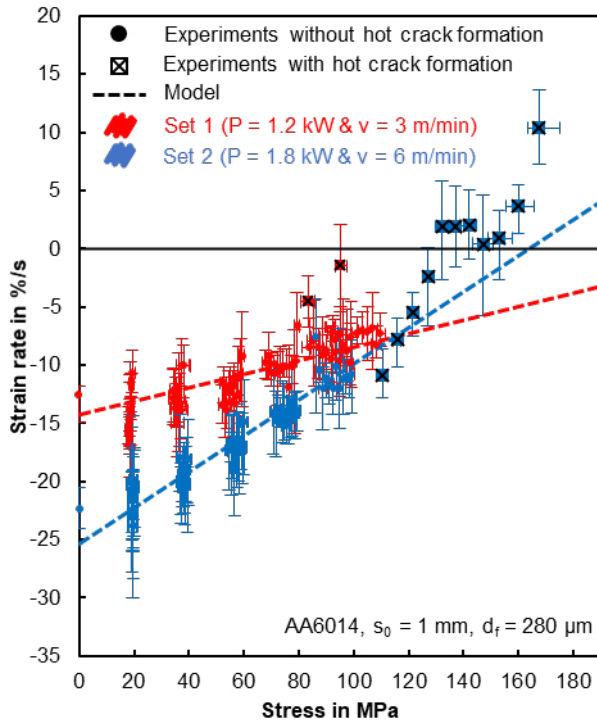


Fig. 4. Comparison of the results gained by the model with the experimental results.

The calculated and the experimental results show that different welding parameters lead to different emerging strain rates for the same acting stress. In absence of hot cracks, the strain rate values emerged higher in the case of welding with parameter setting 2 (blue line in Fig. 4) than in the case of welding with parameter setting 1 (red line in Fig. 4). Since all measured strain rates are in the negative range, this implies that a higher speed and a lower depth specific line energy lead to a higher compression of the solidification zone of the weld.

The slope of the increase of the emerging strain rates appears stronger for parameter setting 2 with the higher velocity of $v = 6$ m/min and the lower depth specific line energy (Set 2, blue) than the slope of the emerging strain rates for parameter setting 1 with the lower velocity of $v = 3$ m/min (Set 1, red). In case of acting stresses beyond 100 MPa this resulted in higher emerging strain rates and in the reliable formation of hot cracks when welding with the higher velocity of $v = 6$ m/min (Set 2, blue).

5. Implications on Process Optimization

The experimental and calculated results indicate that the release of stresses can be optimized by the process parameters.

However, as shown in [3] the process parameters do also influence the resistance of the weld against hot cracking, i.e. the critical strain rate for hot crack formation. In order to access an optimization of the welding parameters, the critical strain rate of the welding process is additionally calculated according to the model presented in [3]. The material properties listed in Table 2 and Table 3 and the solidification path of the AA6014 aluminium alloy Ac170 presented in [14] were used for these calculations.

Table 3. Properties of the solidifying aluminium alloys [2,15]

Material parameter	Abbr.	Value	Unit
Solid fraction of coherency	$f_{s,coh}$	0.95	1
Solidification shrinkage	β	0.065	1
Viscosity	μ	0.001	Pa·s
Critical pressure drop	Δp_c	2000	Pa

The resulting calculated critical strain rate is represented in Fig. 5 by the dashed black curve as a function of the depth specific line energy E_{Depth} . Strain rates, which affect the solidification zone and exceed this critical strain rate lead to the formation of hot cracks.

Furthermore, Fig. 5 presents the emerging strain rates as a function of E_{Depth} calculated according to equation (4) for two exemplary uniaxial tensile stress levels to which the sample was exposed. The dash-dotted curves represent the emerging strain rates in case of welding a sample which is exposed to a uniaxial stress of 60 MPa and the solid curves represent the emerging strain rates in case of welding a sample which was exposed to a uniaxial stress of 110 MPa. The red curves represent the results in case of welding with the parameter setting 1 and the blue curves represent the results in case of welding with the parameter setting 2, as listed in Table 1.

The dotted black line represents the emerging strain rates in case of welding a sample which is not exposed to an external load, i.e. the strain rate $\dot{\epsilon}_{th}$, which results from the cooling of the material as described with equation (3). The dotted curve is valid for both parameter settings.

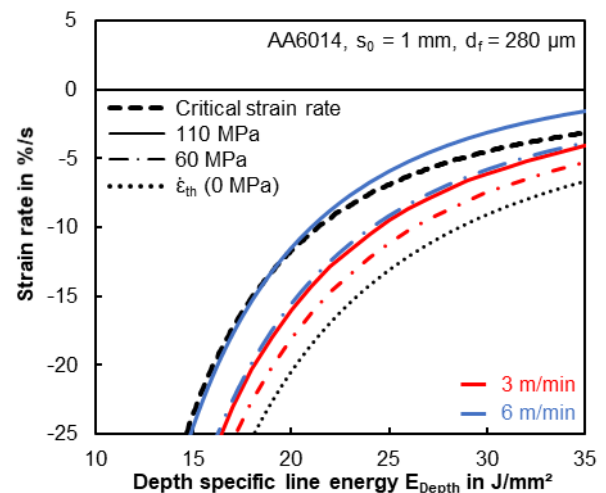


Fig. 5. Calculated hot cracking resistance of laser beam welds in AA6014 and emerging strain rates which affect the solidification zone in case of welding sheets, which are exposed to uniaxial stresses.

The course of the solid blue curve in Fig. 5 shows, that the emerging strain rate exceeds the critical strain rate for a depth specific line energy $E_{Depth} > 18$ J/mm² in case of welding with a velocity of 6 m/min a sheet which is exposed to a uniaxial stress of 110 MPa. As experimentally proven above, this leads to the formation of hot cracks (see blue data points in Fig. 4, which are marked by a black cross).

The emerging strain rates do not exceed the critical strain rate in case of welding the same sheet exposed to the same stress of 110 MPa with a velocity of 3 m/min (red solid curve in Fig. 5). Within the considered range of E_{Depth} the course of the solid red curve is almost congruent to course of the dash-

dotted blue curve. This shows, that the lower velocity of 3 m/min reduces the emerging strain rates in case of 110 MPa almost to the values of the emerging strain rates in case of welding with a velocity of 6 m/min a sheet which is exposed to a uniaxial stress of 60 MPa.

The critical strain rate increases as well as the emerging strain rates with increasing depth specific line energy E_{Depth} . For this reason, the parameter setting 1 with the applied depth specific line energy of $E_{Depth} = 24 \text{ J/mm}^2$ is less susceptible to hot crack formation despite higher strain rates are emerging than in case of welding with the parameter setting 2 with a depth specific line energy of $E_{Depth} = 18 \text{ J/mm}^2$. Towards higher depth specific line energies, the emerging strain rates converge to the critical strain rate, regardless of which welding parameters are applied or which stresses are acting.

This analysis indicates, that in order to achieve a maximum reliability regarding the avoidance of centreline hot cracks in welding sheets which are exposed to tensile stresses, it is beneficial to weld with low welding velocities at depth specific line energies between 20 J/mm^2 and 25 J/mm^2 .

6. Conclusion

In order to understand the influence of residual stresses on the hot crack formation during laser beam welding of aluminium alloys the relation between the acting stresses and the emerging strain rates was investigated experimentally and modelled by analytical equations with sufficient accuracy.

Residual stresses were experimentally simulated by applying uniaxial load to aluminium sheets during welding. The emerging strain rates were measured during welding by means of high-speed digital image correlation. Gained experimental results show a proportional relation between the acting stresses and the emerging strain rates and a good agreement to the analytical model.

The derived analytical model further describes the influence of the process parameters on the emerging strain rates in case of acting stresses. The equations identify the welding velocity and the line energy per depth as the most relevant parameters to reduce the hot cracking susceptibility of such welds.

The analysis shows that it is beneficial to apply low welding velocities and a line energy per depth between 20 J/mm^2 and 25 J/mm^2 in order to achieve maximum reliability regarding the avoidance of hot crack formation in welding sheets, which are loaded by residual stresses.

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